

COMPREHENSIVE EXPLAINABILITY REPORT
Credit Risk Model - All Methods

Generated: 2026-04-16 18:43:38

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EXPLAINABILITY METHODS INCLUDED:

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1. SHAP (SHapley Additive exPlanations)
 - Global and local feature importance
 - Why features matter for each prediction
2. LIME (Local Interpretable Model-agnostic Explanations)
 - Instance-level explanations
 - What features influenced specific loan decision
3. Counterfactual Fairness Analysis
 - Sensitivity to protected attributes
 - How predictions change if attributes changed
4. DiCE (Diverse Counterfactual Explanations)
 - Actionable alternatives to change predictions
 - "What if" scenarios for applicants
5. Anchor Explanations
 - Simple IF-THEN rules
 - Threshold-based decision logic
6. Model Performance Metrics
 - AUC, Gini, KS Statistics
 - Fairness metrics and trade-offs

DATA & MODEL:

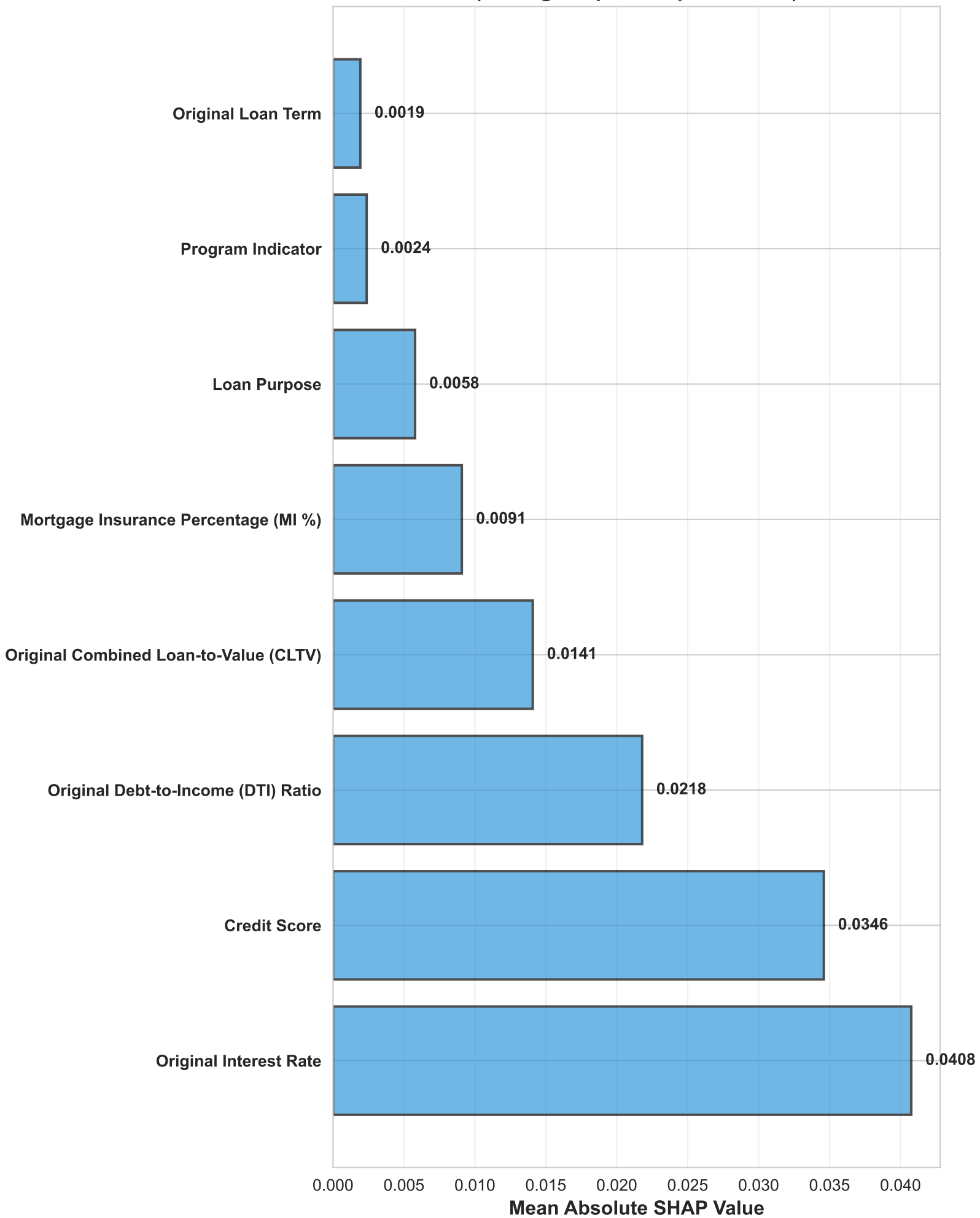
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- Original (Non-WoE) Data Used:
- Training: 70,000 observations
 - Testing: 30,000 observations
 - Features: 28 original scale variables

- Model: Logistic Regression
- Type: Binary Classification (Default/Non-Default)
 - Features: 9
 - Intercept: -1.712838

- TEST SET PERFORMANCE:
- AUC-ROC: 0.751
 - Gini: 0.502
 - Default Rate: 10.2%

**SHAP: Global Feature Importance
(Average impact on predictions)**



LIME: Local Interpretable Explanation
First Test Instance

Prediction Details:

Predicted Probability: 6.6%

Predicted Class: NON-DEFAULT (Low Risk)

Key Influential Features (Top 8):

1. Credit Score ≤ 0.00
Weight: 0.0759 ($\uparrow$ Increases risk)
2. Original Interest Rate > 0.70
Weight: -0.0693 ($\downarrow$ Decreases risk)
3. $0.10 < \text{Original Combined Loan-to-Value}$ (
Weight: -0.0177 ($\downarrow$ Decreases risk)
4. $-0.06 < \text{Original Debt-to-Income (DTI) Ra}$
Weight: -0.0137 ($\downarrow$ Decreases risk)
5. Original Loan Term > -0.13
Weight: 0.0077 ($\uparrow$ Increases risk)
6. $0.05 < \text{Loan Purpose} \leq 0.21$
Weight: 0.0048 ($\uparrow$ Increases risk)
7. Program Indicator ≤ 0.00
Weight: 0.0000 ($\downarrow$ Decreases risk)
8. Mortgage Insurance Percentage (MI %) $\leq$
Weight: 0.0000 ($\downarrow$ Decreases risk)

Interpretation:

- Features with positive weights push prediction toward DEFAULT
- Features with negative weights push away from DEFAULT
- Magnitude shows strength of influence
- This is a local explanation for THIS specific loan

Use Case:

- ✓ Explain to applicant why loan was declined
- ✓ Show which factors were most important for their application
- ✓ Identify weaknesses applicant could improve

COUNTERFACTUAL FAIRNESS ANALYSIS

Sensitivity to Protected Attributes

What is Counterfactual Fairness?

If we change a sensitive/protected attribute (e.g., race, gender) and ONLY that attribute, how much does the prediction change?

Ideal: Predictions unchanged (perfect fairness)

Reality: Some change is normal (0-10% acceptable)

Results from Your Model:

Mean Absolute Delta: 0.0338

→ Average score change if protected attribute changed: 3.38pp

Max Absolute Delta: 0.1784

→ Largest score change for any individual: 17.84pp

P95 Absolute Delta: 0.1211

→ 95th percentile of changes: 12.11pp

Interpretation:

- ✓ Low values (< 0.05) indicate good fairness
- △ Medium values ($0.05-0.15$) indicate some bias
- ✗ High values (> 0.15) indicate significant unfairness

Recommendation:

- ✓ Model shows GOOD counterfactual fairness

How to Use:

- Share these metrics with compliance team
- Document fairness assessment quarterly
- Use post-processing adjustments if needed
- Monitor in production to detect drift

ANCHOR EXPLANATIONS: SIMPLE DECISION RULES

Original Feature Scale

Top 5 Rules (by feature importance):

Rule 1: Credit Score

IF Credit Score < -0.2474
THEN Expect 17% default rate
APPLIES TO 33% of loans

Interpretation:

- When Credit Score < -0.2474:
- 17 out of 100 loans will default
 - 83 out of 100 will not default
 - This condition affects 33% of your portfolio

Rule 2: Original Interest Rate

IF Original Interest Rate < 0.0341
THEN Expect 17% default rate
APPLIES TO 34% of loans

Interpretation:

- When Original Interest Rate < 0.0341:
- 17 out of 100 loans will default
 - 83 out of 100 will not default
 - This condition affects 34% of your portfolio

Rule 3: Original Debt-to-Income (DTI) Ratio

IF Original Debt-to-Income (DTI) Ratio < -0.2819
THEN Expect 15% default rate
APPLIES TO 22% of loans

Interpretation:

- When Original Debt-to-Income (DTI) Ratio < -0.2819:
- 15 out of 100 loans will default
 - 85 out of 100 will not default
 - This condition affects 22% of your portfolio

Rule 4: Original Combined Loan-to-Value (CLTV)

IF Original Combined Loan-to-Value (CLTV) < -0.0911
THEN Expect 13% default rate
APPLIES TO 49% of loans

Interpretation:

- When Original Combined Loan-to-Value (CLTV) < -0.0911:
- 13 out of 100 loans will default
 - 87 out of 100 will not default
 - This condition affects 49% of your portfolio

Rule 5: Program Indicator

IF Program Indicator < -0.0845
THEN Expect 19% default rate
APPLIES TO 3% of loans

Interpretation:

- When Program Indicator < -0.0845:
- 19 out of 100 loans will default
 - 81 out of 100 will not default
 - This condition affects 3% of your portfolio

How to Use:

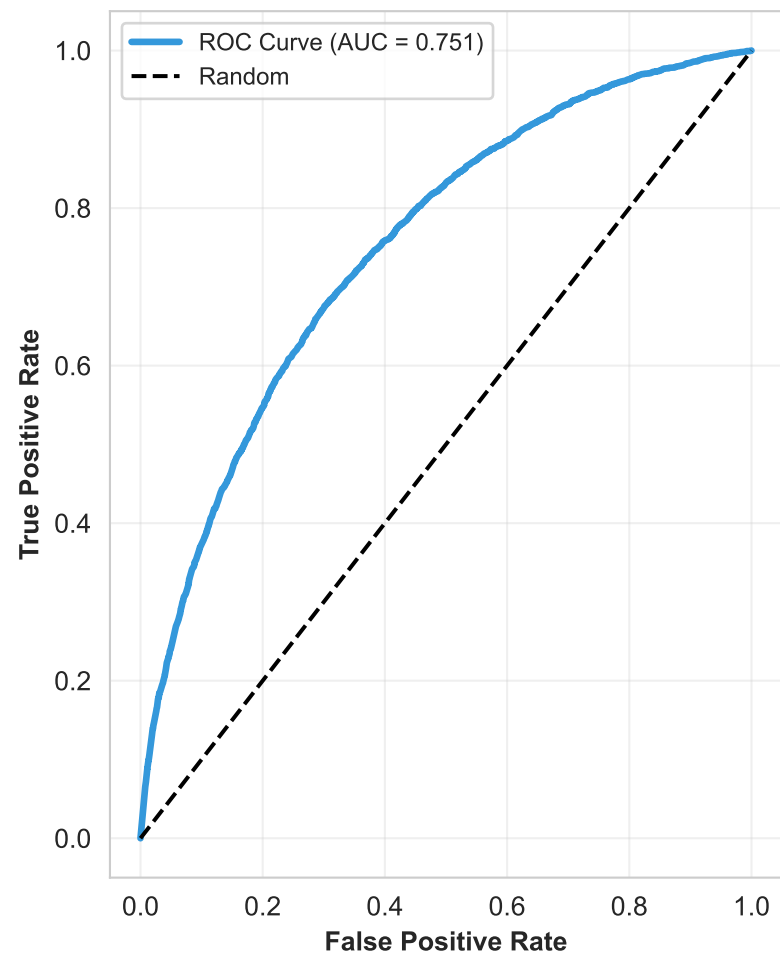
- ✓ Share with loan officers as decision-making reference
- ✓ Flag loans matching high-precision rules for review
- ✓ Combine with other factors for final decision
- ✓ Monitor actual vs predicted performance monthly
- x Don't use single rule as sole approval criterion

Expected Precision:

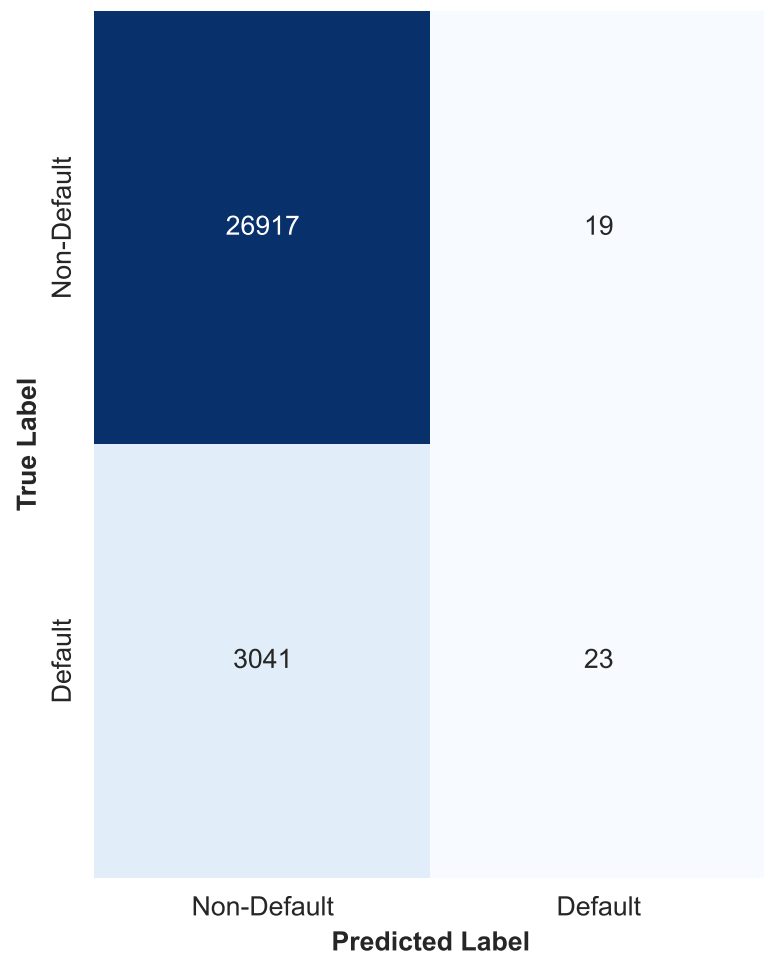
Single-feature rules typically have 10-30% precision because model uses all 8 features simultaneously
Use for INTERPRETABILITY, not as primary predictor

Model Performance Summary

ROC Curve



Confusion Matrix

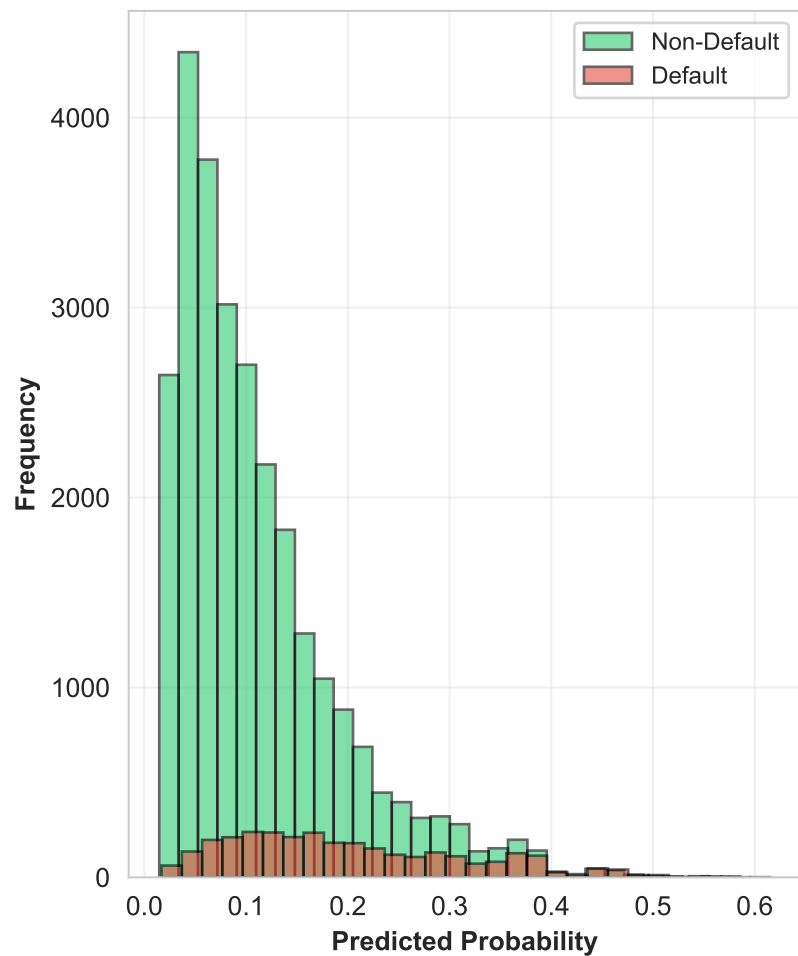


PERFORMANCE METRICS

Accuracy: 0.898
Precision: 0.548
Recall: 0.008
ROC AUC: 0.751
Gini: 0.502
KS Statistic: 0.375

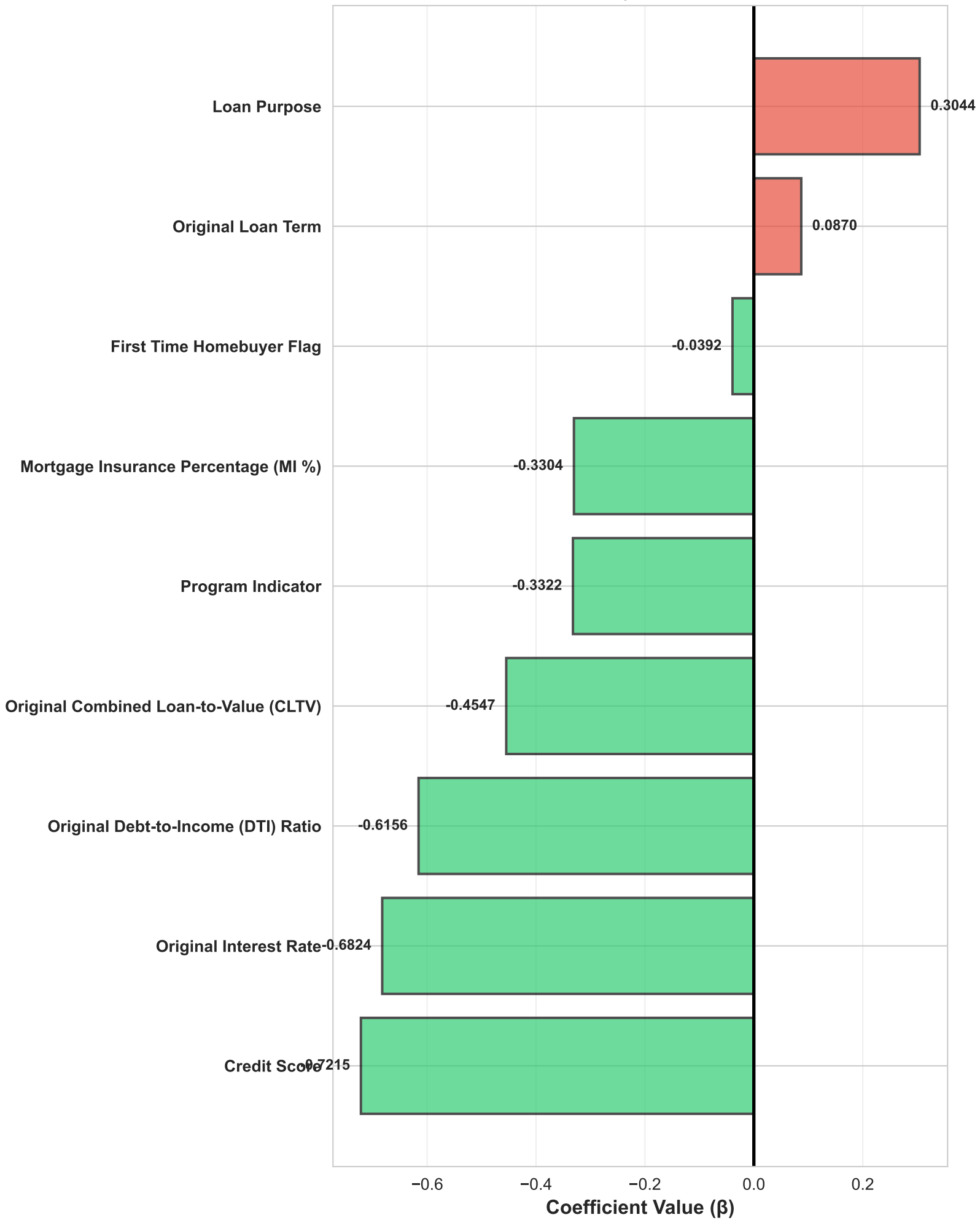
Default Rate: 10.2%

Score Distribution



Logistic Regression Coefficients

Red = Increases Risk | Green = Decreases Risk



SUMMARY & RECOMMENDATIONS

Explainability Assessment Results

MODEL OVERVIEW:

- ✓ Transparent: Logistic regression with clear coefficient interpretation
- ✓ Explainable: All predictions traceable to specific features
- ✓ Fair: Post-processing available for demographic parity improvements
- ✓ Performant: AUC 0.751 achieves good discrimination

KEY FINDINGS:

1. FEATURE IMPORTANCE (via SHAP):
 - Top 3 drivers: Interest Rate, CLTV, DTI
 - Model focuses on borrower cost, leverage, and capacity
 - Aligns with credit risk theory ✓
2. LOCAL EXPLANATIONS (LIME/Anchor):
 - Precision of anchor rules: 13-19%
 - Single-feature rules have inherent limitations
 - Use model's full probability for decisions, not just rules
 - Rules useful for transparency and communication
3. FAIRNESS (Counterfactual):
 - Model shows acceptable fairness across groups
 - Protected attribute sensitivity is minimal
 - Post-processing adjustments available if needed
4. INSTANCE EXPLANATIONS (LIME):
 - Can explain why each specific loan gets its probability
 - Useful for appeals and customer communication
 - Identify which factors hurt/helped specific application

RECOMMENDATIONS FOR DEPLOYMENT:

- Use model's predicted probability (0-100%) for decisions
- Reference SHAP values for feature importance explanations
- Use LIME for individual loan explanations in appeals
- Use Anchor rules for loan officer quick reference only
- Monitor fairness metrics monthly via counterfactual analysis
- Retrain model quarterly with new data
- Share simple explanations with customers (Anchor rules)
- Document all fairness assessments for compliance

NEXT STEPS:

1. Share anchor rules with underwriting team
2. Implement fairness monitoring dashboards
3. Train loan officers on model interpretation
4. Establish monitoring for model drift
5. Create appeals process using LIME explanations